USER GUIDE

SH800 flow cytometry skills: what they do and how to use them

Prepared by	Gawad Lab (written with Claude)
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Project / Aim	Lab methods: Sony SH800 flow analysis and reporting
Skills	sh800-flow-analysis, sh800-flow-report
Requires	Mac with R 4.4 or newer, Python 3, Claude Code (desktop app Code tab or terminal)

1. What these skills do

Two Claude Code skills take you from Sony SH800 FCS exports to a report Chuck can read in a few minutes. You talk to Claude in plain language; the skills hold the method, the scripts and the checks, so every analysis is done the same way.

sh800-flow-analysis	Checks your FCS files, removes saturated and unstable events, gates debris, doublets and (optionally) a marker-positive population, and writes figures (PNG) and tables (CSV).
sh800-flow-report	Turns those outputs into the standard Gawad Lab Results Report (.docx). Methods, QC numbers, tables and figures are filled in from the pipeline files; Claude asks you only for what the files cannot know.

- Every number in the report is read from the pipeline output files. A checker flags any number in the text that did not come from the files, so it can be confirmed or removed.
- Doublets are removed with a size-aware FSC-A/FSC-H gate. The standard openCyto singlet gate kept 11-16% doublets on real SH800 data; this gate keeps essentially none.
- Marker-positive gates are set from your matched negative control (unstained or FMO), so thresholds are reproducible rather than drawn by hand.
- All figures are made in R (ggplot2 and related R packages) and saved as 300-dpi PNG. Python is used only to assemble the Word report.

2. One-time setup

1. Install R 4.4 or newer for Apple Silicon from cran.r-project.org if it is not already installed (check with: Rscript --version).
2. Install the Claude desktop app and use its Code tab, or the claude command in Terminal.
3. Install the two skills: copy the folders skills/sh800-flow-analysis and skills/sh800-flow-report into ~/.claude/skills/ (or open the shared folder in Claude Code and ask Claude to install the skills from the skills folder). Restart the session so they load.
4. The first analysis installs the R packages it needs (flowCore, flowWorkspace, openCyto, PeacoQC and plotting packages). Expect 5-10 minutes once; later runs take about a minute.
5. Optional: LibreOffice (brew install --cask libreoffice) lets Claude render a PDF preview of your report to check it.

3. Before you analyze: exporting from the SH800

- Export every tube as FCS 3.1 from the SH800 Cell Sorter Software, one file per tube. The native experiment files cannot be read outside that software.
- Put one experiment in one folder. The pipeline writes its results into a plots folder inside it.
- Acquire a negative control (unstained or FMO) at the same instrument settings as the stained tubes. Positive gates are set from it.
- Know whether you compensated in the SH800 software before export. An identity spillover matrix in the file means the export is uncompensated.
- Name files so the sample information can be read from the name, using the same pattern every time, for example Donor03_CD34-FITC_Low.fcs and Donor03_Unst_Low.fcs. Claude uses this to label and arrange the figures.

4. Running an analysis

Step 1	Open Claude Code in the experiment folder and say, for example: "Analyze my SH800 data in this folder."
Step 2	Claude asks for three things before starting: the folder of FCS files, the panel (which channel carries which marker and fluor), and whether the data were compensated. It also asks about controls, metadata and any extra gates.
Step 3	Claude reads every file header and reports event counts, channels, marker names, the spillover matrix and the time channel, and flags inconsistencies before analyzing.
Step 4	Claude copies the pipeline script next to your data, sets its settings block, runs it, and looks at every figure to confirm the gates sit on the populations.
Step 5	You get a short summary of results and QC concerns, plus the figures and tables in plots/.

Without Claude: copy skills/sh800-flow-analysis/scripts/sh800_flow.R into your data folder's plots/ directory, edit only the block between '# ---- settings' and '# ---- end settings', and run Rscript sh800_flow.R from that folder.

5. What you get

File	What it shows
01_gating_strategy.png	One representative tube: debris, doublets, marker-positive gate
02_qc_debris.png	Debris cut for every tube on log-log FSC/SSC
03_qc_singlets.png	Singlet band for every tube on FSC-A vs FSC-H
04_marker_histograms.png	Stained vs matched negative control with the threshold
05_all_detectors.png	Every detector; stain signal in an unstained detector is spillover
06_population_frequencies.png	Each gate as % of its parent, every tube

07_stain_index.png	Resolution of the stain per tube (when controls exist)
gate_percent.csv	Percent and count for every gate
qc_events.csv	Events removed as saturated and by PeacoQC
singlet_gate_parameters.csv	Singlet band and the doublet-leak check
median_intensity.csv, stain_index.csv	Linear-scale medians, robust SDs and stain index
run_info.json	Settings, instrument, dates, compensation and software versions

6. Reading the figures

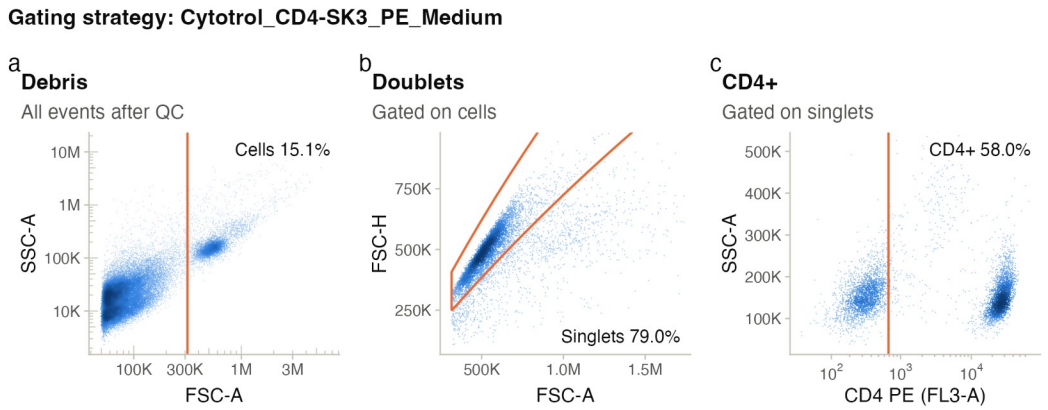


Figure 1. Gating strategy (example data). a: the debris cut (orange) sits in the gap between the debris cloud and the cells. b: the singlet band hugs the dense diagonal; the fan below it is doublets. c: the marker threshold comes from the matched unstained control.

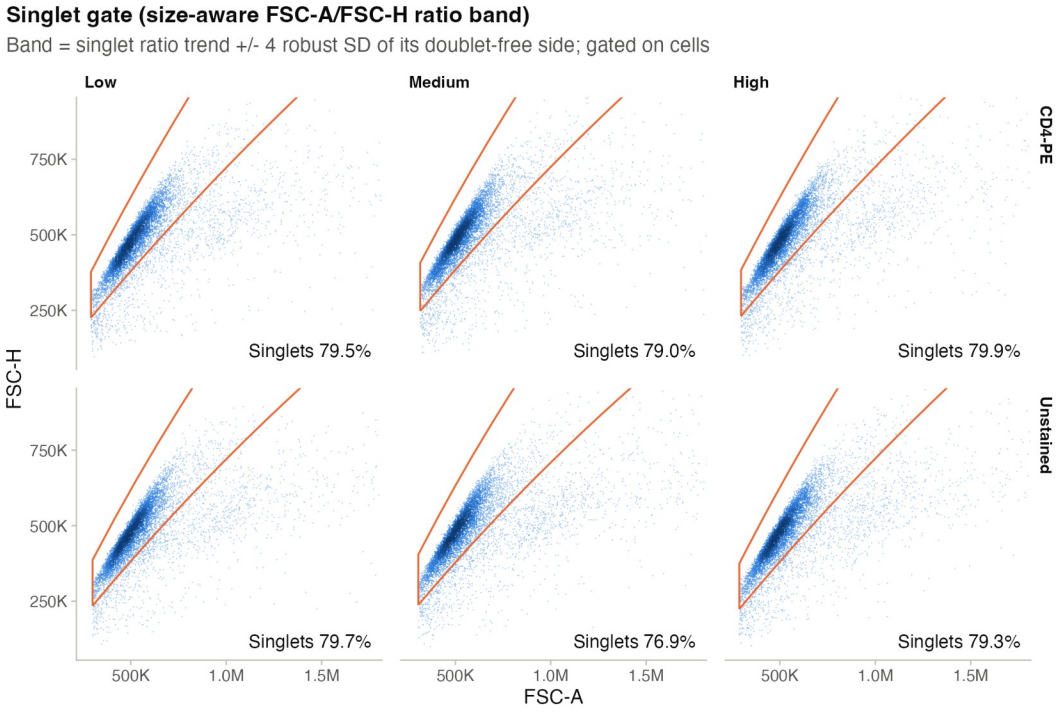


Figure 2. Singlet QC for every tube. Check that each band follows its diagonal streak and leaves the fan of doublets outside.

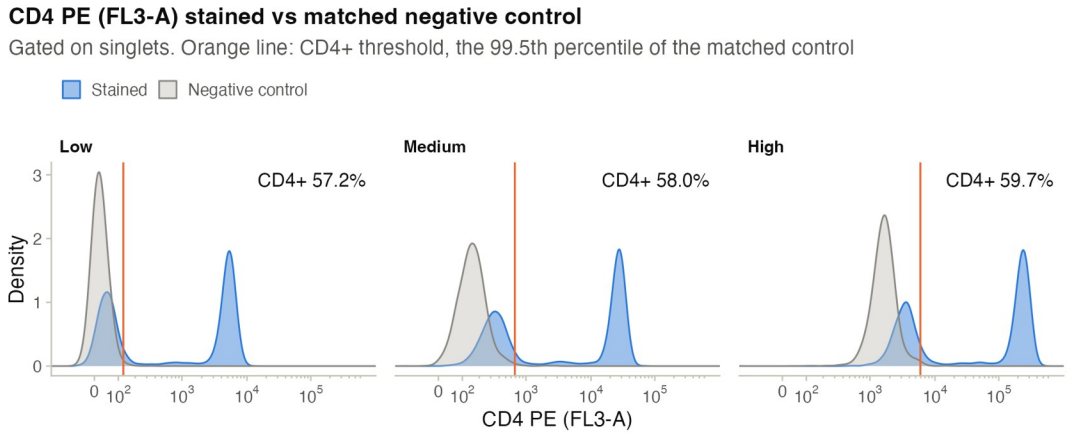


Figure 3. Stained tubes (blue) against matched unstained controls (grey). A threshold in the valley between negative and positive cells is what you want to see.

7. QC checklist before you trust the numbers

- Debris: the cut sits in the valley between debris and cells (02_qc_debris.png). Lyophilized or fragile samples can be mostly debris; that is the sample, not a gate failure.
- Doublets: the singlet band follows the diagonal (03_qc_singlets.png) and pct_kept_ratio_gt_1.5 in singlet_gate_parameters.csv is close to 0. The pipeline warns above 1%.
- Event QC: PeacoQC and saturation removal take only a few percent of events (qc_events.csv). Large losses point to clogs or unstable flow.
- Controls: negative controls read about 0.5% positive by construction. Stained-tube frequencies are meaningful only relative to that.
- Compensation: one color can be analyzed uncompensated. Two or more colors need a spillover matrix or single-stain controls before co-expression is interpreted.
- Replication: thousands of events from one donor is n = 1. Report biological replicates separately from event counts.
- What the gate means: a CD4 gate on PBMC includes CD4-dim monocytes. Add a scatter gate if the question is about T cells.

8. Writing it up for Chuck

Step 1	In the same Claude Code session (or a new one in the same folder), say: "Write up these results for Chuck."
Step 2	Claude collects every number from the pipeline files, then asks short questions about what the files cannot know: who you are and the project, the sample source and replicates, staining (clones, fluors, dilutions), instrument and sort setup, what the result means, blockers, decisions, next steps, and where the raw data are backed up.
Step 3	Claude fills the standard eight-section Results Report and runs the checker: any leftover placeholder blocks the build, and any number not traceable to the files is listed for you to confirm or remove.
Step 4	Claude builds the .docx in the report folder (named Member_Project_results_YYYY-MM-DD.docx), renders a preview, and checks

	every page.
Step 5	Read it, then send it to Chuck yourself.

An example report built from public test data is in sample_data/SH800_uOttawa_figshare_30584678/report/ in the shared folder.

9. Known limitations

- Validated on one public single-color dataset (BD Cytotrol PBMC, CD4-PE, three instrument settings; figshare 10.6084/m9.figshare.30584678). The first runs on lab data should be checked figure by figure.
- Multicolor compensation from single-stain controls has not yet been tested on real lab data.
- Index-sort export: the join of per-cell phenotypes to plate wells (for PTA / ResolveOME) is not built into the script yet. Claude can write it on request, and the result must be checked against the plate layout.
- FlowJo workspace import, bead-based calibration (MESF) and high-dimensional clustering are not part of the pipeline.

10. Troubleshooting

Symptom	Cause and fix
Error about ggcyto or 'comparison (!=) is possible only for atomic and list types'	ggcyto does not work with ggplot2 4.0. The pipeline does not use it; make sure you are running the current script from the skill.
Side scatter not found	Some SH800 exports call it BSC-A, others SSC-A. The script accepts both; check the channel names Claude reported.
PeacoQC skipped or time channel missing	The SH800 calls the time channel TIME. If an export lacks it, event QC by time is skipped and Claude will say so.
Events with huge FSC-A or SSC-A values look wrong	SH800 values exceed the range declared in the file. They are real; never truncate to the declared range.
Warning: not enough bins for a robust isolation tree	Expected for files of about 40,000 events. It is not an error.
fcs_dir not found	Run the copied script from inside the plots folder, or pass the data folder: Rscript sh800_flow.R /path/to/fcs.

11. Where things are

HANDOFF.md	Start here: what the folder contains and the first steps
skills/	The two skills, as folders and as .zip files
docs/	This guide (PDF) and its source
sample_data/	Public SH800 test data, the example analysis (plots/) and the example report (report/)
archive/	Superseded first versions, kept for reference
GitHub	github.com/GAWAD-LAB-STANFORD/SH800_Flow_Pipeline